Homework 5
(20 points)

Homework 5-1

Customer

Customer ID	First Name	Surname	Telephone Number
123	Robert	Ingram	555-861-2025
456	Jane	Wright	555-403-1659 555-776-4100
789	Maria	Fernandez	555-808-9633

a. Convert this relation into 1NF.

Customer(CustomerID, First Name, Surname, Telephone Number)

Employees' Skills

Employee	Skill	Current Work Location
Brown	Light Cleaning	73 Industrial Way
Brown	Typing	73 Industrial Way
Harrison	Light Cleaning	73 Industrial Way
Jones	Shorthand	114 Main Street
Jones	Typing	114 Main Street
Jones	Whittling	114 Main Street

b. Convert this relation into 2NF.

Employees'Skills(Employee, Skill, Current Work Location)

c. Convert this relation into 3NF.

Tournament Winners(<u>Tournament</u>, <u>Year</u>, Winner, Winner Date of Birth)

Tournament Winners

<u>Tournament</u>	<u>Year</u>	Winner	Winner Date of Birth
Indiana Invitational	1998	Al Fredrickson	21 July 1975
Cleveland Open	1999	Bob Albertson	28 September 1968
Des Moines Masters	1999	Al Fredrickson	21 July 1975
Indiana Invitational	1999	Chip Masterson	14 March 1977

Homework 5-2

Why this table in not in 1NF?
Convert it into 1NF.

Product ID	Color	Price
1	Red, Green	10
2	Yellow	20
3	Green	15
4	Yellow, Blue	17
5	Red	22

Product Table

Homework 5-3

Why this table in not in 2NF?
Convert it into 2NF.

CustID	StoreID	Location
1	1	LA
1	3	SF
2	1	LA
3	2	NY
4	3	SF

Purchase Table

Homework 5-4

Why this table in not in 3NF?
Convert it into 3NF.

BookId	GenreID	GenreType	Price
1	1	Science	20
2	2	Sports	35
3	1	Science	17
4	3	Movie	26
5	2	Sports	31

Book Table

Homework 5-5

Convert into 1NF, 2NF and 3NF

(You do not need to show filled table)

It is for your easiness.

Project Code	Project Title	Project Manager	Project Budget	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
PC010	Pensions System	M Phillips	24500	S10001	A Smith	L004	IT	22.00
PC010	Pensions System	M Phillips	24500	S10030	L Jones	L023	Pensions	18.50
PC010	Pensions System	M Phillips	24500	S21010	P Lewis	L004	IT	21.00
PC045	Salaries System	H Martin	17400	S10010	B Jones	L004	IT	21.75
PC045	Salaries System	H Martin	17400	S10001	A Smith	L004	IT	18.00
PC045	Salaries System	H Martin	17400	S31002	T Gilbert	L028	Database	25.50
PC045	Salaries System	H Martin	17400	S13210	W Richards	L008	Salary	17.00
PC064	HR System	K Lewis	12250	S31002	T Gilbert	L028	Database	23.25
PC064	HR System	K Lewis	12250	S21010	P Lewis	L004	IT	17.50
PC064	HR System	K Lewis	12250	S10034	B James	L009	HR	16.50
PC010	Pensions System	M Phillips	24500	S10001	A Smith	L004	IT	22.00
				S10030	L Jones	L023	Pensions	18.50
				S21010	P Lewis	L004	IT	21.00
PC045	Salaries System	H Martin	17400	S10010	B Jones	L004	IT	21.75
				S10001	A Smith	L004	IT	18.00
				S31002	T Gilbert	L028	Database	25.50
				S13210	W Richards	L008	Salary	17.00
PC064	HR System	K Lewis	12250	S31002	T Gilbert	L028	Database	23.25
				S21010	P Lewis	L004	IT	17.50
				S10034	B James	L009	HR	16.50